

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Designed and implemented **distributed system architectures** for enterprise Java Spring Boot microservices supporting **large-scale infrastructure** across multiple cloud regions, handling **10M+ daily transactions** with sub-100ms latency requirements using **AWS EC2, S3, Lambda** and container orchestration via **Docker**.
- Developed **C++ performance-critical components** for real-time data processing pipelines, optimizing **network packet processing** and **distributed computing workflows** with **advanced data structures and algorithms**, reducing end-to-end latency by **40%**.
- Led design reviews and code reviews for cross-functional teams building **distributed network systems**, establishing best practices around **system design, testability, efficiency, and style guidelines**; mentored **5+ junior engineers** on large-scale infrastructure patterns and **compute technologies**.
- Built end-to-end **data pipelines and ETL workflows** using **Python, SQL (PostgreSQL, MySQL, Oracle)** for operational analytics and monitoring dashboards; architected **data storage solutions** supporting real-time query performance across **petabyte-scale datasets** with **25% improvement** in query throughput.
- Triageed and resolved **P1/P2 production incidents** in distributed payments and ERP infrastructure using **Splunk, Dynatrace, CloudWatch** for root-cause analysis; implemented automated alerting and **hardware/network/service diagnostics** reducing MTTR by **35%**.
- Deployed and maintained **CI/CD infrastructure** using **Jenkins, Git, GitHub Actions** for automated testing and release management; integrated hardware deployment pipelines with infrastructure-as-code practices, ensuring reliability across **100+ production services**.

## SKILLS

**Languages & System Development** – C++, Java, Python, SQL, Bash/Shell scripting.

**Distributed Systems & Infrastructure** – distributed computing, large-scale system design, network architecture, compute technologies, storage systems, hardware architecture, microservices.

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, container orchestration, Jenkins, GitHub Actions, Git, CI/CD pipelines.

**Data Engineering & Analytics** – data pipelines, ETL workflows, data structures, algorithms, PostgreSQL, MySQL, Oracle, pandas, NumPy, scikit-learn, data warehousing.

**Monitoring & Reliability** – Splunk, Dynatrace, production incident management, system diagnostics, SLA management, alerting frameworks.

**Software Design & Collaboration** – REST API development, code review, design reviews, system integration, testing, accessibility standards, technical documentation.

**Frameworks & Tools** – Spring Boot, React, Angular, Power BI, Tableau, Flask, FastAPI, Linux administration.

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of  
Ireland, Dublin**

**02/2026**

## **PROJECTS**

### **Distributed Payments Operations Dashboard (HCL Technologies)**

- Architected a **large-scale distributed system** supporting real-time SWIFT messaging and payments transaction processing for a tier-1 banking client, handling **50M+ daily payment messages** across geographically distributed compute nodes with **99.99% uptime SLA**.
- Implemented **data structures and algorithms** for efficient message routing, deduplication, and state management across **distributed network infrastructure**; optimized **network packet processing** to achieve **sub-50ms end-to-end latency** for payment settlement.
- Designed **system monitoring and diagnostics** using **Splunk and Dynatrace** to track hardware, network, and service operations; built operational dashboards for **ISO 20022 message flows** and **cross-border payments**, enabling **100% audit trail compliance** and regulatory reporting.

### **Ireland Jobs Tracker – AI-Powered Resume Generation Engine (Personal Project)**

- Built a full-stack distributed application orchestrating **large-scale data pipelines** using **Apify scrapers, Claude API integration, and reportlab PDF generation**, processing **500+ job descriptions daily** with **99.2% accuracy** in structured data extraction.
- Engineered **algorithm-driven prompt optimization** for AI job description analysis, implementing constraint guardrails and JSON-structured data validation; deployed **compute-optimized workflows** via **GitHub Actions CI/CD** and **Vercel infrastructure**.
- Designed and tested **system reliability features** including retry logic, error handling, and distributed task queuing to ensure **99%+ success rate** across asynchronous resume generation; documented system architecture and API contracts for maintainability.

## **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)